Assignment 2:
Text Classification via Naïve Bayes

Topics: Conditional Probability, Laplace Smoothing, Text Vectorization, Model Evaluation

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Objective In this assignment, you will explore how text representation techniques (feature extraction) interact with different probabilistic distribution assumptions in Naïve Bayes classifiers. You will implement and evaluate these models from scratch to predict spam messages.

Dataset setup

- **Dataset source:** SMS Spam Collection Dataset (or IMDb Movie Reviews)
- **Data splitting:** Split the dataset into training and test sets using an 80/20 ratio (`train` $= 0.8$, `test` $= 0.2$).
- **Preprocessing:** Apply standard text preprocessing techniques, including lowercasing, punctuation removal, stripping special characters, optional stop-word filtering, and tokenizing into words.

Task instructions

You must construct three distinct model pipelines by pairing specific f**eature extraction** methods with their corresponding **Naïve Bayes formulations**:

1. **Model A (Multinomial without smoothing):**
   - *Feature extraction:* Bag-of-Words / Term frequency (counts total occurrences per word f_w).
   - *Model:* Multinomial Naïve Bayes with **no smoothing** ($\alpha = 0$).
2. **Model B (Multinomial with Laplace smoothing):**
   - *Feature extraction:* Bag-of-Words / Term frequency (f_w).
   - *Model:* Multinomial Naïve Bayes with **Add-One Laplace smoothing** ($\alpha = 1$).
3. **Model C (Bernoulli with Laplace smoothing):**
   - *Feature extraction:* Binary / One-Hot encoding (presence/absence indicator $y_w \in \{0, 1\}$).
   - *Model:* Bernoulli Naïve Bayes with **Add-One Laplace smoothing** ($\alpha = 1$).

Evaluation criteria & required deliverables

Your submission will be assessed on four core pillars:

1. Implementation

- Build the vocabulary index and calculate prior probabilities $P(Y)$ and class-conditional probabilities $P(X_i|Y)$ from scratch.
- Implement prediction mechanisms using log-probabilities ($\log P(X|Y)$) to prevent numerical underflow issues.

2. Quantitative & qualitative evaluation

- **Quantitative metrics:** Report Precision, Recall, F_1-Score, and Accuracy across all three models in a clean comparative summary table.
- **Qualitative analysis:** Identify and analyze 3–5 false positives and false negatives from the test set. Explain *why* the classifier misclassified these specific texts.

3. Visualization

- **Confusion matrices:** Plot side-by-side confusion matrices for Models A, B, and C on the test set.
- **Top features:** Visualize the top 10 words with the highest log likelihood ratio $\log \frac{P(w|\text{Spam})}{P(w|\text{Ham})}$ to identify the strongest predictive indicators.

4. Discussion & critical analysis

- **The zero probability problem:** Discuss how Model A handles unseen vocabulary words in the test set compared to Model B.
- **Independence assumption violation:** Analyze a sentence where the "naïve" conditional independence assumption fails (e.g., phrase pairs like "not bad" vs. "bad") and describe how it impacts model output.
- **Multinomial vs. Bernoulli:** Compare the trade-offs of counting term frequency versus binary occurrence for text datasets of varying document lengths.